

Kryptos, the trifid cipher, and automated analysis of classical cryptosystems

Seth of CryptoCrap pointed out that Sanborn had said that he is an “[arithemah](#)” and finished high school mathematics outside of school. So, when he says “matrix”, he is unlikely to be referring to a mathematical matrix as Bauer et al wrote; he is probably just referring to a grid arrangement of characters. This is also apparent from [his talk at Big Tech Day](#) in June 2013, where he describes K1-K3 as being “matrix codes”.

Scheidt has previously described the last section as having 98 characters, which would mean that it begins “70BK” and thus has 27 different characters. Among classical cryptosystems, the most well-known system with 27 different characters is the “[Trifid cryptosystem](#)” invented by Felix Delastelle. It is likely that K4 is based on a known classical cryptosystem as the Kryptos sculpture reflects the history of cryptography and codes.

Thomas at [Klaus's Krypto Kolumne](#) wrote that the references to “ROWS” and “LAYER TWO” in the two different decipherments of K2 also are likely to indicate Trifid. One more possibility is that “ID BY ROWS” is a reference to “TRIFID BY ROWS”; or at least some kind of “multifid” cipher. It could be either each block is encrypted one of the rows in the sculpture, or a variant of trifid where the plaintext is written in by rows and the ciphertext is read out by columns. Bifid is less likely because the K4 ciphertext has 26 characters, not 25, and digrafid is an ACA invention c1960.

The [Kryptools website](#) goes through the public information, and suggests possible grid sizes – the K1 and K2 grid paper has 31 columns, while the K3 grids are 42x8 and 14x24.

Since Kryptos was installed in 1990, several papers have been published on automated cryptanalysis of classical ciphers, starting with Safavi-Naini et al's papers around 1993 using simulated annealing. Gillogly described “shotgun hill climbing” in an ACA article in 1995 (also used as the main technique in the 2008 book “[Decoding the IRA](#)”). Clark wrote his [PhD thesis](#) on this topic in 1998, Cowan compared hill climbing and simulated annealing for many ciphers including trifid between 2007 and 2010, and Lasry published his [PhD thesis](#) in 2017 referencing all of these. It is perhaps possible that “HILL” found on the sculpture could refer to “hill climbing” as a cryptanalysis technique. Gillogly used this technique to solve K1-K3.

If we are trying to find the key to a trifid cipher by brute force, having 11 characters of known plaintext (“BERLINCLOCK”) may reduce the search space from $27! (1.1 \times 10^{28})$ by a factor of 3^{23} – to as low as 2.0×10^{12} – not yet accounting for symmetry. For a given position (e.g. the first character in mapping) we only need to test six possibilities due to symmetry, so this reduces the space by another 26/6 factor. Thus, for a particular period it's even possible to do a brute force search. It is not generally necessary though as simulated annealing is a very good technique for finding the key, assuming we know the plaintext language.

The [ACA guidelines](#) for the period length for a trifid are 5-15 with a ciphertext length of 120-150 letters.

I examined all period lengths from 2 to 51, with all possible rotations of the ciphertext, and both variants of trifid (inserting plaintext in the matrix “by columns” and “by rows”) but didn't find anything.

More obscure ideas

It is interesting that the row length of the last three sculpture rows is 31. Sanborn had said that his 2010 hints were “three-pronged” i.e. crib text, website, and grid sheet release, and another time at a dinner he said that there was information to be gleaned from the grid sheets themselves. Certainly we can now see that each row has 31 characters. Perhaps “DYAHR” is referring to DAY plus HR and is thus 24 plus 7, and we can use a trifid variant with alternating period lengths of 24 and 7, or 7 and 24, starting at “Q70BK”, “70BK”, “0BK” or “UOXO”.

If the ciphertext length is 99, the factors are 3, 9, 11, and 33.

If it's 100 (e.g. add in YAR, hopefully at the beginning or end, not sprinkled through it somewhere) we have 2, 4, 5, 10, 20, 25, 50. If it's 96 (leave out a letter somewhere) we have many factors – 2, 3, 4, 6, 8, 12, 16, 24, 32 and 48.

However, none of these ideas led anywhere.

We have to be careful with the scoring function using logged n-gram frequencies – should # (the 27th character) be scored like an X, like a question mark, a full stop, an E, or a space? Also, with the known plaintext, we can allow for scoring which gives texts which are just close to the crib, without matching it exactly; in case the encryption has a mistake in it.

Based on Michael Cowan's old site cryptoden – we can also use “tries” to change the scoring function so that it's based on the number of complete words found in the decipherment – which is mentioned in his description of the “[Churn algorithm](#)”.

The example given was “Junk Jewelfy jingles, jolts jerky jackanapes. Jealous jades jot jabberwocky, jerboa jumps jaguarundi” which is not really plain English. Unfortunately, the simulated annealing algorithm is quite slow to assess scores for this compared to summing logged n-gram frequencies.

Fortunately, the K4 plaintext seems unlikely to be this obscure as Sanborn had said it is “plain English” (unless there are lots of #s or Xs sprinkled everywhere).

Finally, as with the Hill cipher – in Trifid, we could use two different alphabets, one for enciphering and one for deciphering. However, there is not then enough ciphertext to determine what keys were used – the [unicity distance](#) would be around 60 characters. The only real likely option would be using two of the sculpture alphabets.

Appendix – timeline

- 1983 paper – The technique of “Simulated annealing” is first described in *Science* by [Kirkpatrick, Gelatt, and Vecchi](#)
- 1986 Glover writes about “[random restart hillclimbing](#)”, which is another name for “shotgun hill climbing”. In the various articles, it isn't always clear whether the algorithm is “first ascent” (move to the first neighbour you find with a better score) or “steepest ascent” (explore all neighbours and move to the one with the best score).
- 1988-1990 Kryptos codes are developed and installed
- 1993 Forsyth and Safavi-Naini “[Automated cryptanalysis of substitution ciphers](#)” (simulated annealing)
- 1994 Giddy and Safavi-Naini “[Automated cryptanalysis of transposition ciphers](#)” (simulated annealing)
- 1995 “Shotgun hillclimbing” is described by Gillogly – The Cryptogram Nov/Dec 1995 issue
- 1998 Andrew J Clark OUP “[Optimization heuristics for cryptology](#)”, chapter 3 concerns attacks on classical ciphers with simulated annealing, genetic algorithms and tabu search.
- 2007 Michael J Cowan – articles in The Cryptogram – M12007, SO2007 on Simulated annealing. Example code is at

<http://www.cryptogram.org/resource-area/computer-column-programs/simulated-annealing-example-using-playfair/>

<http://www.cryptogram.org/resource-area/computer-column-programs/aristo-excel-worksheet-and-playfair-simulated-annealing-program/>

<http://practicalcryptography.com/cryptanalysis/stochastic-searching/cryptanalysis-bifid-cipher/#c-code-for-breaking-bifid> (James Lyons)

Cowan wrote:

Trifid can also be solved by SA, which again finds the solution much more quickly than hillclimbing. There are 6 possible mixed alphabets for any particular Trifid key, depending on assignment of numbers in the tableau. I've illustrated this below using just the first 3 columns:

```
POW POW POW POW POW POW
111 111 222 111 222 333
111 111 222 111 222 333
123 132 213 321 231 312
```

These six different assignments produce six variations in mixed alphabet, all of which give the same encipherment or decipherment, shown below. It can be seen that 3-letter groups move position in the mixed alphabet, and also the letters within a group change in order. In the simulated annealing program it pays to include swaps that move these groups around.

```
POWERANDMIGHTBCFJKLQSUUVXYZ#
PWONMDEARLSQY#ZUXVIHGFKJTCTB
BTCGIHJFKREADPNMNVUXQLSZY#
#ZYXVUSQLKJFCBTHGIMNAREWOP
#YZSLQXUVWMDNPOAEKFKJHIGCTB
BCTJKFGHIVXUZ#YQSLRAEDMNOWP
```

- 2008 Michael J Cowan publishes articles on “Solving Trifid” with simulated annealing in The Cryptogram (May/June 2008, Computer Corner, page 12)
- 2008 Cowan – “[Breaking short playfair ciphers with the simulated annealing algorithm](#)” (Cryptologia). Mentions SA works well on bifid, and better than hill climbing.
- 2012 [McLaughlin thesis](#) – section 3.5 - see particularly section 3.5.2 on SA parameter choices
- 2018 Lasry thesis “[A methodology for the cryptanalysis of classical ciphers with search metaheuristics](#)” is published online, discussing the references above.